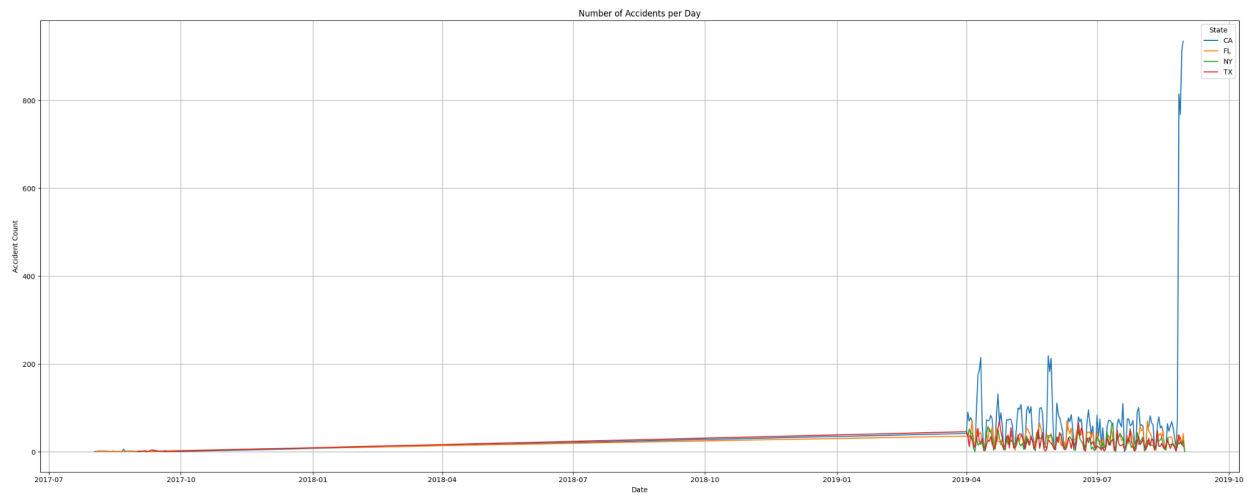
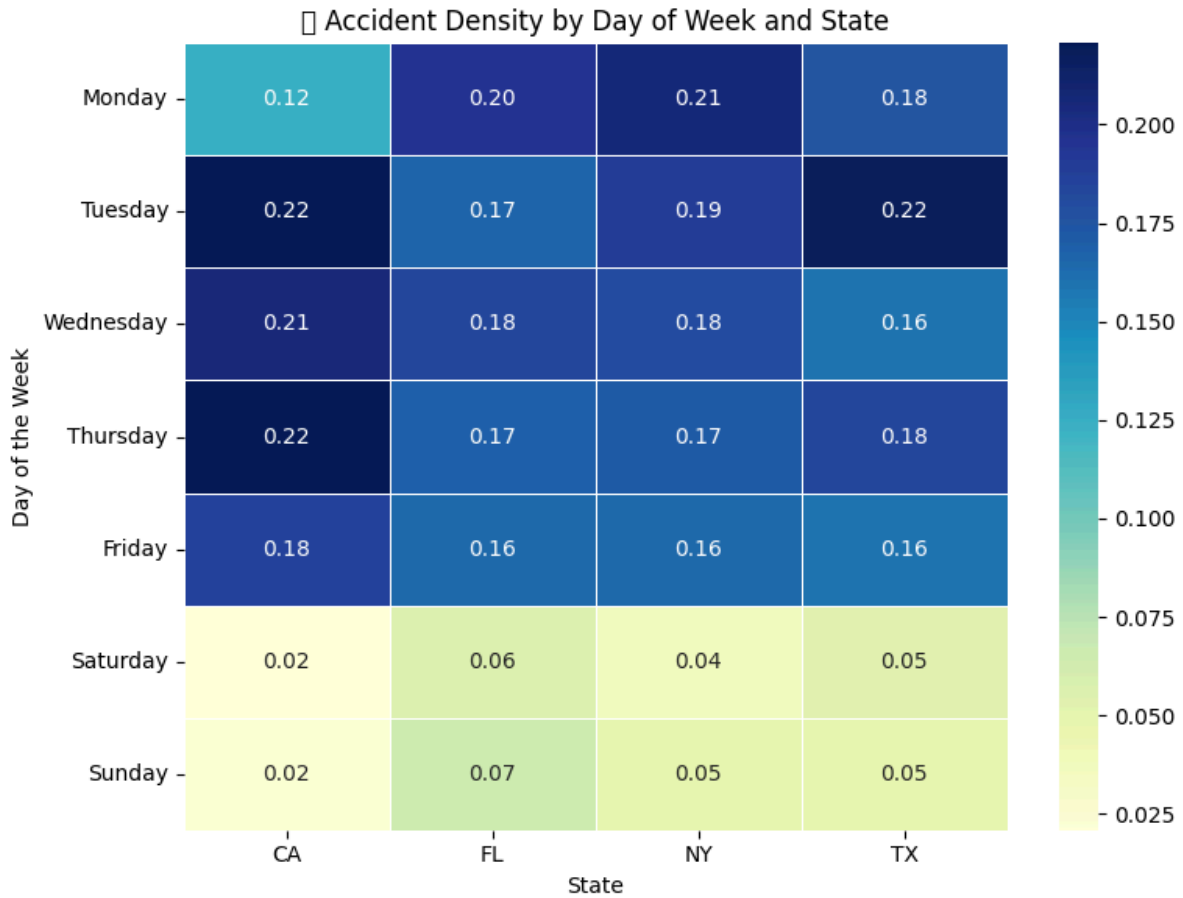


Homework 5



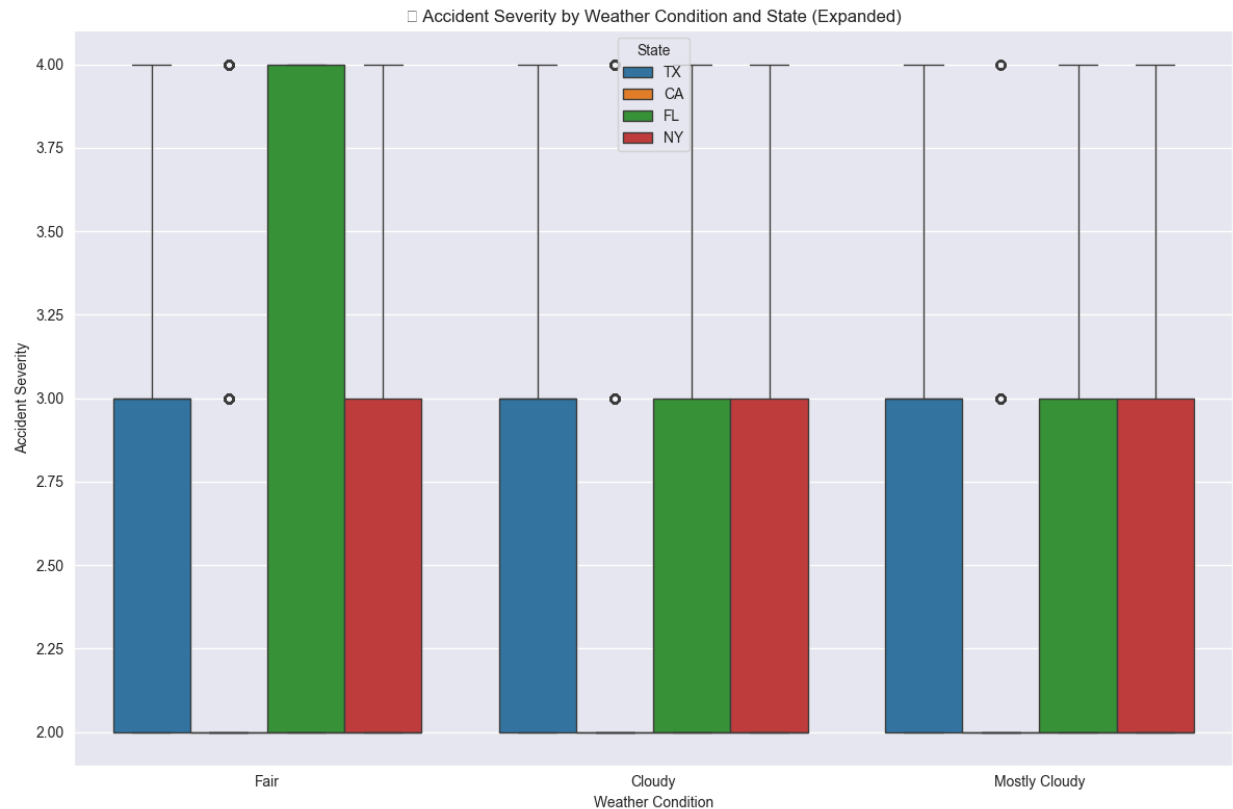
Task 1 Insights

- Accident frequency is highly variable day to day.
- **Texas (TX)** shows a consistently high number of accidents, peaking more frequently than other states
- **California (CA)** and **Florida (FL)** follow similar trends, with fluctuations but lower daily counts compared to TX.
- **New York (NY)** generally shows fewer daily accidents, likely due to dataset size or driving patterns.
- Overall, weekdays tend to have higher accident counts, indicating commuter traffic influence.



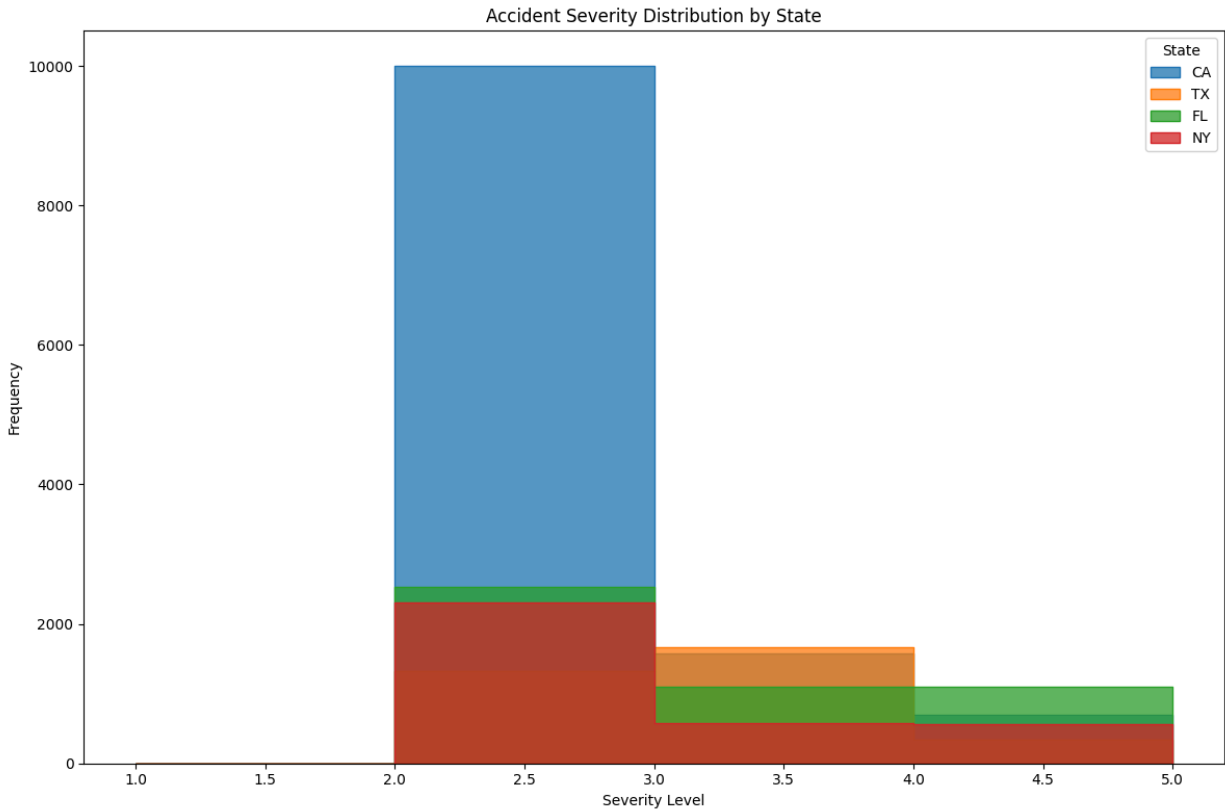
Task 2 Insights

- **Texas (TX)** shows consistent high accident density across weekdays.
- **California (CA)** and **Florida (FL)** have relatively uniform density across the week, slightly lower on weekends.
- **New York (NY)** shows a mild increase during weekdays, hinting at urban commuting patterns.
- Weekdays overall see higher density, confirming weekday traffic is a major accident contributor.



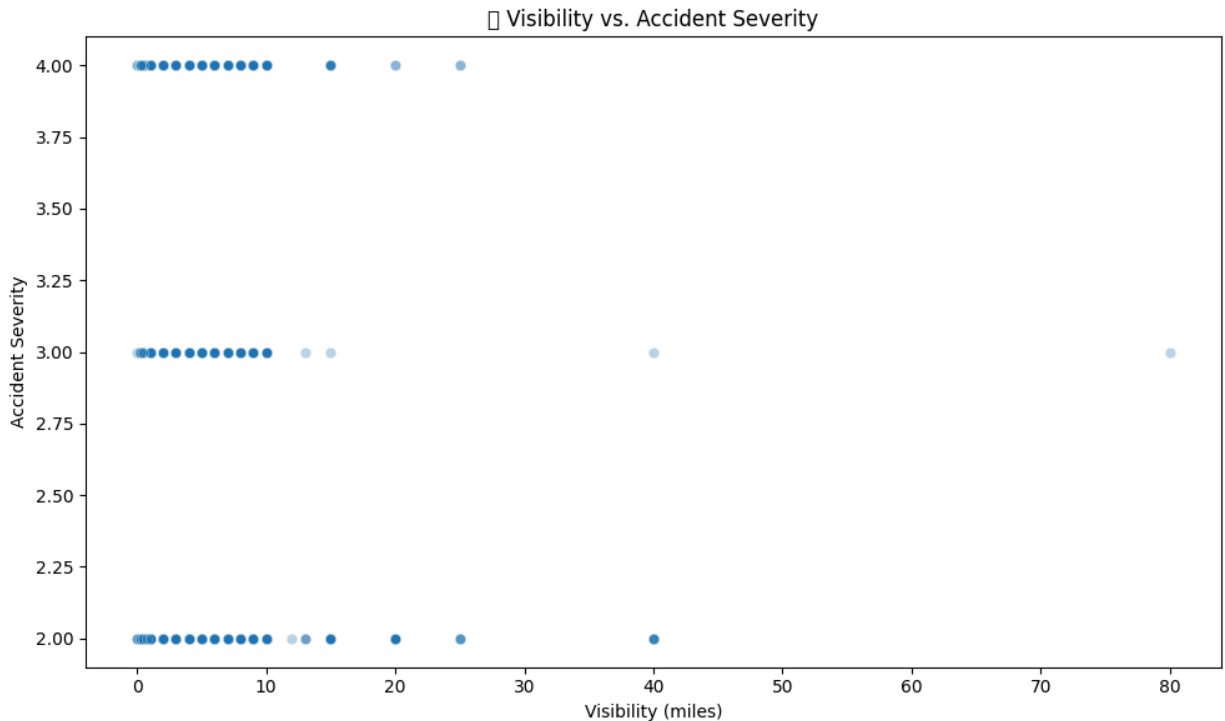
Task 3 Insights

- **Severity mostly remains between level 2 and 3** across all states and weather conditions.
- **Fair weather** surprisingly shows accidents of similar severity to Cloudy conditions — suggesting human factors, not just weather, play a role.
- Minor variation between states, but no strong correlation between weather condition and severity in this sample.
- Implication: Even good weather does not eliminate the risk of moderately severe accidents.



Task 4 Insights

- **Severity Level 2 dominates** across all states, indicating most accidents are moderately minor.
- Very few cases in Severity Level 4, showing that severe accidents are less common.
- **Texas (TX)** has a slightly higher frequency of severe accidents compared to others.
- Data suggests that while accidents are frequent, most are not high severity.



Task 5:

a) Question Raised:

Does lower visibility increase accident severity?

b) Choice of Plot:

- **Plot type:** Scatter plot
- **Reason:** To visualize the relationship between visibility distance (in miles) and accident severity as continuous variables.
- Scatter plot helps identify any pattern or trend — especially to see if lower visibility correlates with higher severity.

Insights:

- Overall, **there is no strong correlation** between visibility and accident severity in this dataset.
- **Slight trend:** Lower visibility (below 5 miles) shows a mild increase in higher severity accidents.
- However, many accidents still happen in high-visibility conditions, suggesting other contributing factors (e.g., distraction, speed).

- Implication: Reduced visibility is a risk factor, but not the sole determinant of accident severity.